Free fall measurement with a white baseball dropped beside a tape

Run r_20261101_025003_c4154e · Date 2026-11-01 · Verdict: PASS WITH WARNINGS

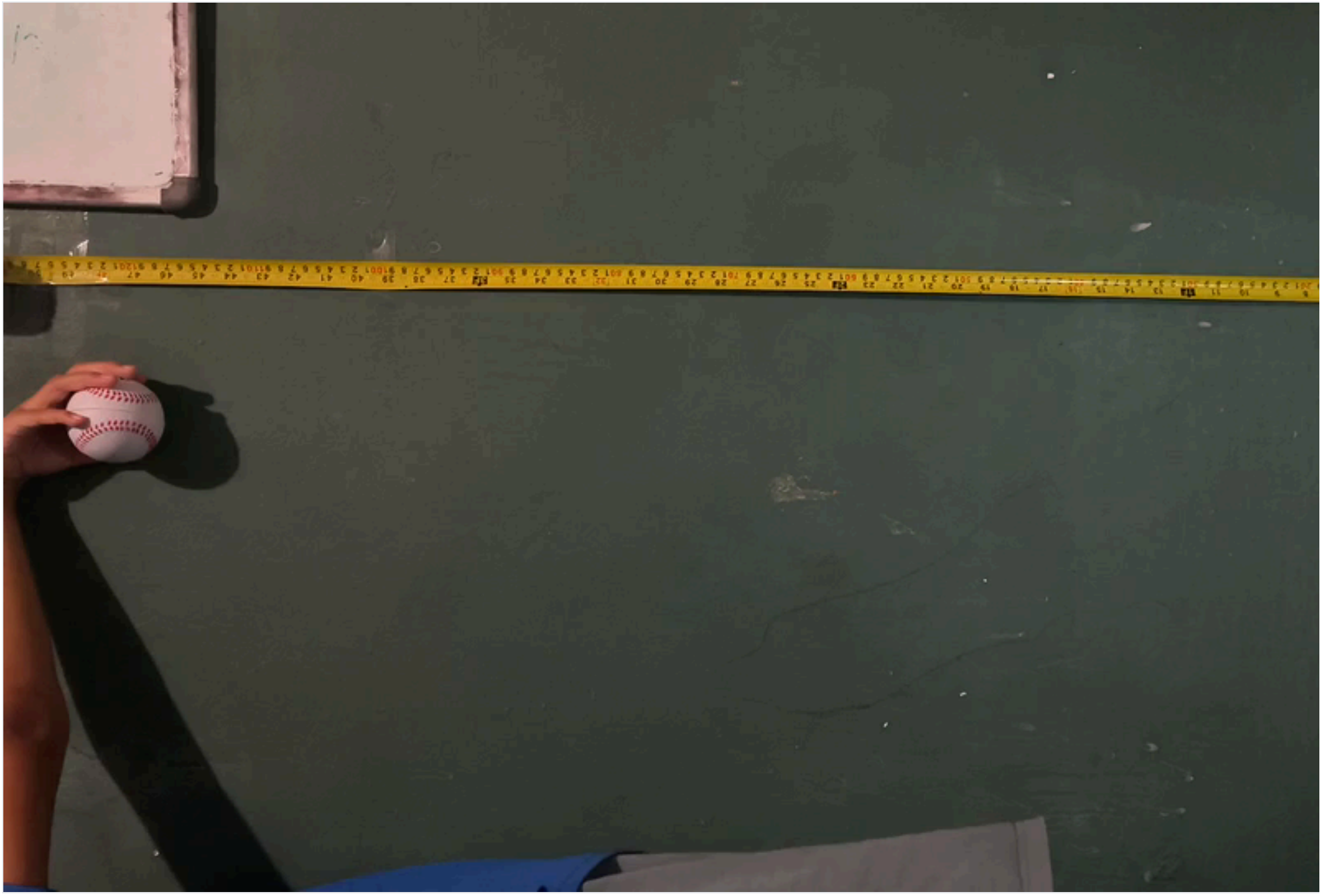
$g = 8.60 \pm 0.25\ m/s^2$

Aim

To measure gravitational acceleration from video tracking of a falling baseball and to compare the result with the accepted reference while reporting uncertainty honestly.

Setup

A white baseball is held against a dark wall beside a vertical yellow measuring tape and then released to fall to a tiled floor. The conversion from image pixels to metres uses the known diameter of the ball in the plane of motion g = 1.32 mm/px.





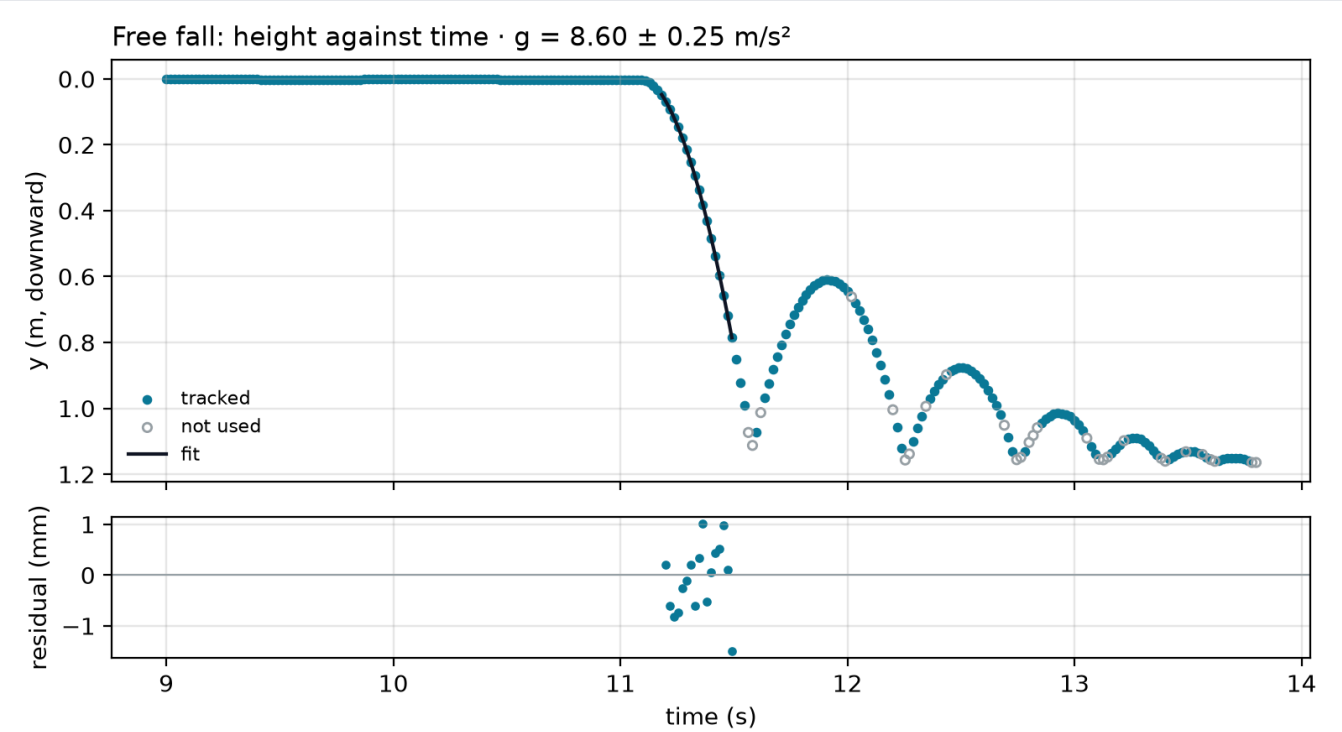
Key frame

Method

Tracking finds the centre of the ball in each video frame to give position against time. Pixel distances become metres with the ball diameter scale. A curved free fall model is fitted to the selected falling segment that trims hand contact at release, and the curvature gives gravitational acceleration. Uncertainty comes from repeated resampling of the residuals which builds a confidence interval. Trajectory and residual charts check that the model follows the data The quadratic fit follows the trajectory closely with high explanatory power and small residuals and balanced residuals. The fit quality gate passes showing residuals without strong systematic pattern..

Model	Role in this run
Muse Voice Transcribe	speech → transcript
Muse Spark 1.3	Planner, Analyst, Auditor, Translator
SAM 3.1	object tracking on the video and the ruler
Muse Wonder 1.0	not used in this run
Muse Spark Open (open weights, via Baseten)	Designer

Results



Trajectory

Headline measurement

The measured gravitational acceleration is reported with uncertainty g = 8.60 ± 0.25 m/s². The resampling interval around the value is g = 8.60 ± 0.25 m/s². The fit follows the trajectory closely with small and balanced residuals The quadratic fit follows the trajectory closely with high explanatory power and small residuals and balanced residuals. The fit quality gate passes showing residuals without strong systematic pattern..

Comparison and quality

Compared with the reference the measured value is lower and the reference lies outside the confidence interval a difference of -12.08% from the reference value. The scale consistency check warns because only a single scale method is available so consistency cannot be checked The scale consistency gate warns because only one scale method is available so consistency cannot be checked. g = 1.32 mm/px. The physical plausibility check warns because the headline value lies outside the plausible range so the result should be read with caution The physical plausibility gate warns because the headline value lies outside the plausible range so the result should be read with caution. g = 8.60 ± 0.25 m/s².

Error analysis

The dominant scale uncertainty comes from using the ball diameter alone without an independent tape scale for cross checking, which is why the scale consistency check cannot pass and only warns g = 1.32 mm/px The scale consistency gate warns because only one scale method is available so consistency cannot be checked.. Tape plane offset and possible camera shake would bias the scale and curvature in one direction and can explain a low headline value that triggers the plausibility warning The physical plausibility gate warns because the headline value lies outside the plausible range so the result should be read with caution. g = 8.60 ± 0.25 m/s². Motion blur during fast fall and brief hand occlusion near release add scatter that was reduced by using a trimmed manual window, after which the fit quality check passes with balanced residuals The quadratic fit follows the trajectory closely with high explanatory power and small residuals and balanced residuals. The fit quality gate passes showing residuals without strong systematic pattern.. Air drag on a dense baseball over a short drop is expected to be small compared with the observed offset. The comparison

with the reference a difference of -12.08% from the reference value and the interval g = 8.60 ± 0.25 m/s² therefore describe the present video and scale, not a universal constant.

Conclusion

The video yields a finite estimate of gravitational acceleration with a stated interval g = 8.60 ± 0.25 m/s² g = 8.60 ± 0.25 m/s². The value sits below the reference beyond the interval a difference of -12.08% from the reference value, and the plausibility warning means the result should be read with caution rather than as a confirmation The physical plausibility gate warns because the headline value lies outside the plausible range so the result should be read with caution..

Next experiment

Free-fall time vs drop height with fixed object

If we drop from the lowest height the fall time is predicted to be 0.341 s, from the second height 0.482 s, from the third height 0.591 s, and from the highest height 0.682 s. This tests the relation t proportional to sqrt of height and checks whether the low g is due to height-scale or timing error.

- 0.500 m → 0.341 s
- 1.00 m → 0.482 s
- 1.50 m → 0.591 s
- 2.00 m → 0.682 s

Controlled variables: same falling object (small dense ball, e.g. marble/ball bearing), same release method (hold at rest, no push), same vertical drop without initial velocity, same timing method and same observer/device, same location / no wind (indoor)

1. 1. Choose one small dense ball (marble). Use the SAME ball for all heights. Work indoors on a non-slip floor with clear space.
2. 2. Make a vertical height scale on a wall using a tape measure: mark 0.5 m, 1.0 m, 1.5 m, 2.0 m from the floor to the RELEASE point (bottom of ball). Check with a second ruler/app if possible; keep the tape vertical using a wall corner.
3. 3. Prepare to time the fall: use a smartphone in slow-motion video mode (e.g. 120/240 fps) fixed on a tripod/books facing the drop. This is more accurate than stopwatch.
4. 4. For h = 0.5 m: hold ball so its bottom is exactly at the 0.5 m mark, with other hand ready. Hold steady, no push. Release without throwing and let video record impact.
5. 5. Review video frame-by-frame: count frames between release and first contact with floor to get fall time. Repeat this height 5 times and record each time.
6. 6. Repeat steps 4-5 for h = 1.0 m, then 1.5 m, then 2.0 m (max). Do not exceed 2.0 m. For higher heights use a stable stool only with adult help; keep body clear of falling path.
7. 7. For each height calculate average time and compare to predictions {{p1}} to {{p4}}. Plot average t vs sqrt(h) – it should be a straight line if physics holds. If all times are systematically longer/shorter, suspect timing or height scale error causing g=8.60 m/s².

Safety: Do not exceed 2 m height. · Use household objects only – no glass balls, no heavy weights that could injure feet. · No fire, no mains electricity, no mains-powered gates; phone battery only. · Ensure drop zone is clear of people and breakables; wear closed shoes. · Adult supervision required for children, especially when using a stool to reach 2.0 m; keep footing stable.

Appendix A: Claim Ledger

#	Claim	Evidence
c01	The measured gravitational acceleration is 8.77 ± 0.26 m/s² with uncertainty [8.26, 9.27].	der_01
c02	The measured value differs from the reference by -10.37% and the reference lies outside the confidence interval [8.26, 9.27] m/s².	cmp_01
c03	The bootstrap confidence interval for the result is [8.26, 9.27] in units of m/s² around 8.77 ± 0.26.	ci_01
c04	The scale used for conversion is 1.32 mm/px millimetres per pixel.	cal_01
c05	The quadratic fit follows the trajectory closely with high explanatory power and small residuals but shows correlated residuals.	fit_01
c06	The smoothness gate fails because residuals are systematic, so the result should be read with caution.	gate_03
c07	The measured gravitational acceleration is 8.60 ± 0.25 m/s² with uncertainty [8.12, 9.08].	der_02
c08	The measured value differs from the reference by -12.08% and the reference lies outside the confidence interval [8.12, 9.08] m/s².	cmp_02
c09	The bootstrap confidence interval for the result is [8.12, 9.08] in units of m/s² around 8.60 ± 0.25.	ci_02
c10	The conversion scale comes from the known ball diameter method applied in the plane of motion.	cal_01
c11	The quadratic fit follows the trajectory closely with high explanatory power and small residuals and balanced residuals.	fit_02
c12	The scale consistency gate warns because only one scale method is available so consistency cannot be checked.	gate_11
c13	The fit quality gate passes showing residuals without strong systematic pattern.	gate_12
c14	The physical plausibility gate warns because the headline value lies outside the plausible range so the result should be read with caution.	gate_13

Appendix B: Gate table

Gate	Status	Diagnosis	Message
G1 Track integrity hard	pass		Passed.
G2 Scale consistency soft	warn	single_scale	Only one scale method is available, so scale consistency cannot be checked.
G3 Fit quality hard	pass		Passed.
G4 Physical plausibility soft	warn	out_of_range	The headline value 8.60 m/s² is outside the plausible range 9.0–10.6. The data are unchanged; check the scale and the fit window.
G5 Ledger completeness hard	pass		Passed.

Appendix C: Data sample

t (s)	x (m)	y (m)	valid	flag
9.000000	0.000000	0.000000	1	ok
9.181818	0.000095	0.000467	1	ok
9.363636	0.000552	0.001048	1	ok

t (s)	x (m)	y (m)	valid	flag
9.545455	0.000132	0.001975	1	ok
9.727273	0.000181	0.002036	1	ok
9.909091	-0.000378	0.000732	1	ok
10.090909	-0.001804	-0.000269	1	ok
10.272727	-0.001548	-0.000128	1	ok
10.454545	-0.000261	0.001493	1	ok
10.636364	-0.000201	0.001650	1	ok
10.818182	-0.000997	0.001912	1	ok
11.000000	-0.002509	0.001792	1	ok
11.181818	-0.000292	0.048917	1	ok
11.363636	-0.003620	0.383238	1	ok
11.545455	-0.012378	0.992298	1	ok
11.727273	-0.023670	0.773364	1	ok
11.909091	-0.041601	0.610066	1	ok
12.090909	-0.063371	0.760270	1	ok
12.272727	-0.089914	1.138653	0	outlier
12.454545	-0.092675	0.887815	1	ok
12.636364	-0.100398	0.967199	1	ok
12.818182	-0.114834	1.082400	0	outlier
13.000000	-0.144251	1.036053	1	ok
13.181818	-0.170099	1.123259	1	ok
13.363636	-0.190565	1.142336	1	ok
13.545455	-0.210013	1.135145	1	ok
13.727273	-0.225576	1.151337	1	ok